

Milad Khademi Nori

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RESEARCH

Theory-first continual learning on foundation models. My doctoral work built a mathematical framework for class-incremental learning (the Infeasibility and Feasibility Theorems, NeurIPS 2024) and the hybrid-replay methods that follow from it for single-machine and federated settings (ICML 2025, ICLR 2025). My current program asks why pretrained backbones evade the infeasibility result and answers with geometry: an exponential bound on task confusion in the Mahalanobis margin of the representation, a closed-form account of the bias that regularized softmax heads acquire when trained in stages, and theory-derived heads, prompts and adapters. Its forward edge is continual learning for agents, where memory consolidation is formalized as an online rent-or-buy problem. Every result in the program was produced on a single GPU. Earlier work: communication-efficient federated learning (IEEE Transactions on Communications, 110 citations) and energy-harvesting sensor networks.

Keywords: continual and class-incremental learning; federated learning; foundation models and representation geometry; learning theory; continual learning agents.

EDUCATION

Ph.D., Electrical and Computer Engineering (Machine Learning), Queen's University, Kingston, Canada 2019–2024

All degree requirements completed January 2024. Thesis: Federated Learning and Incremental Learning: Private Non-IID Learning. Supervisor: Prof. Il-Min Kim. Doctoral research funded by Huawei Technologies Canada.

M.Sc., Electrical Engineering, Amirkabir University of Technology, Tehran, Iran Sep 2016–Jul 2019

Ranked first in the class. Thesis: an energy-efficient routing protocol for energy-harvesting wireless sensor networks in the Internet of Things. Supervisor: Prof. Saeed Sharifian.

B.Sc., Electrical Engineering (Digital Electronic Systems), Semnan University, Semnan, Iran Sep 2012–Jun 2016

Ranked first in the class. Project: design and implementation of a Bluetooth-controlled quadcopter, with a team of engineers. Supervisor: Prof. Naser Eskandarian.

APPOINTMENTS

Postdoctoral Fellow, Department of Computer Science, Toronto Metropolitan University Mar 2024–present

Supervisor: Prof. Guanghui (Richard) Wang. From September 2026, a one-year research position supported jointly by Prof. Wang's group and the group of Prof. Il-Min Kim (Queen's University).

- Built the geometric theory of class-incremental learning on pretrained backbones: two papers accepted

at NeurIPS 2026, four manuscripts under review at AAAI 2027, and a slate of manuscripts on memory consolidation in continual agents prepared for ICLR 2027.

Research Assistant, Queen’s University (Innovation Park), Kingston

Jan 2023–Apr 2024

With Prof. Sidney Givigi. Exploratory analysis of prostate-cancer datasets across modalities (CT volumes, mass spectrometry): data cleaning and preprocessing pipelines, volumetric data handling, landmark registration; four technical reports and the accompanying code.

Research Assistant, Huawei-funded doctoral research program, Queen’s University with Huawei Technologies Canada (Ottawa)

Aug 2019–Nov 2023

Research on non-IID deep learning: communication burden and data heterogeneity in federated learning; catastrophic forgetting and task confusion in continual learning. Semi-annual in-person results presentations to Huawei; five papers in top venues came out of the program.

Instructor, Lab Instructor and Teaching Assistant, Department of Electrical and Computer Engineering, Queen’s University

Sep 2019–Sep 2023

Taught, ran laboratories and mentored: MATLAB programming and mathematics for about 110 undergraduate engineering students per week; discrete-time signal processing in Python for about 90 students.

PUBLICATIONS

Google Scholar: 180 citations as of September 2026. First author on every paper listed.

Peer-reviewed conference papers

1. **M. Khademi Nori**, G. Wang. A Mahalanobis margin $\gamma_{\min}$ bound on task confusion in pretrained class-incremental learning: from infeasibility to exponential attenuation. *Advances in Neural Information Processing Systems 39 (NeurIPS)*, 2026. **Accepted (poster)**.
2. **M. Khademi Nori**, G. Wang. The structural bias of ℓ_2 -regularized cross-entropy heads in class-incremental learning: characterization, limits, and a gauge-anchoring fix. *Advances in Neural Information Processing Systems 39 (NeurIPS)*, 2026. **Accepted (poster)**.
3. **M. Khademi Nori**, I.-M. Kim, G. Wang. Autoencoder-based hybrid replay for class-incremental learning. *International Conference on Machine Learning (ICML)*, PMLR vol. 267, 2025.
4. **M. Khademi Nori**, I.-M. Kim, G. Wang. Federated class-incremental learning: a hybrid approach using latent exemplars and data-free techniques to address local and global forgetting. *International Conference on Learning Representations (ICLR)*, pp. 84398–84411, 2025.
5. **M. Khademi Nori**, I.-M. Kim. Task confusion and catastrophic forgetting in class-incremental learning: a mathematical framework for discriminative and generative modelings. *Advances in Neural Information Processing Systems 37 (NeurIPS)*, pp. 47678–47707, 2024.

Journal articles

6. **M. Khademi Nori**, Y. Ge, I.-M. Kim. On the effectiveness of activation noise in both training and inference for generative classifiers. *IEEE Access*, vol. 11, pp. 131623–131638, 2023.
7. **M. Khademi Nori**, S. Yun, I.-M. Kim. Fast federated learning by balancing communication trade-offs. *IEEE Transactions on Communications*, vol. 69, no. 8, pp. 5168–5182, 2021. 110 citations.
8. **M. Khademi Nori**, S. Sharifian. EDMARA2: a hierarchical routing protocol for EH-WSNs. *Wireless Networks*, vol. 26, no. 6, pp. 4303–4317, 2020.

Other peer-reviewed contributions

9. **M. Khademi Nori**, S. Sharifian. A QGSA cluster head selection approach for hierarchical routing protocol in the EH-WSNs. *4th Iranian Conference on Signal Processing and Intelligent Systems (ICSPIS)*, IEEE, 2018.

Manuscripts under review

10. **M. Khademi Nori**, G. Wang. Joint rehearsal restores the global gauge and leaves one scalar ν : a closed-form theory of replay in class-incremental learning. Under review, *AAAI 2027*.
11. **M. Khademi Nori**, G. Wang. Two biases, one offset ω : long-tailed class-incremental learning in closed form. Under review, *AAAI 2027*.
12. **M. Khademi Nori**, G. Wang. Predicting class-incremental difficulty before training: a neural-collapse diagnostic for backbone selection. Under review, *AAAI 2027*.
13. **M. Khademi Nori**, G. Wang. The margin as a control variable: damped backbone fine-tuning for class-incremental learning. Under review, *AAAI 2027*.

Manuscripts in preparation (ICLR 2027) [confirm: which to list, and author lists]

14. **M. Khademi Nori** [fill: co-authors]. Promotion is rent-or-buy: exact ledgers and competitive limits for memory consolidation in continual agents.
15. **M. Khademi Nori** [fill: co-authors]. Anchoring buys geometry-independence: competitive analysis of batched memory consolidation.
16. **M. Khademi Nori** [fill: co-authors]. Hysteresis in memory consolidation: optimal promotion and demotion policies for continual learning agents.
17. **M. Khademi Nori**, I.-M. Kim, G. Wang. Hybrid representative autoencoder-based replay for federated incremental semantic segmentation.
18. **M. Khademi Nori**, I.-M. Kim, G. Wang. UniFDIL: unified federated domain- and class-incremental learning. [confirm: final title]

Thesis

19. **M. Khademi Nori**. Federated Learning and Incremental Learning: Private Non-IID Learning. Ph.D. thesis, Queen's University, 2024.

AWARDS AND HONOURS

- Excellent Reviewer, IEEE Transactions on Network Science and Engineering, IEEE Communications Society, 2023.
- Dean's Research Fund, Travel for Postdoctoral Fellows, Faculty of Science, Toronto Metropolitan University, 2025 (to present at ICML 2025, Vancouver).
- Huawei-funded doctoral research program, Queen's University, 2019–2023.
- Ranked first in the class, M.Sc., Amirkabir University of Technology, 2019.
- Ranked first in the class, B.Sc., Semnan University, 2016.
- Huawei Certified Network Associate (HCNA), 2017. [confirm: year]

PROFESSIONAL SERVICE

Reviewing. Reviewer for ICML, NeurIPS, ICLR, CVPR and other top-tier conferences since 2021 (six years), including ICML 2026, NeurIPS 2026 and AAAI 2027. Journal reviewer for IEEE Transactions on Communications, IEEE/ACM Transactions on Networking, IEEE Transactions on Network Science and Engineering and IEEE Transactions on Pattern Analysis and Machine Intelligence.

Community. Vice-President, speaker and public-relations chair, Ingenuity Lab, Queen’s University, 2019–2023: regular talks on the broader impacts and opportunities of artificial intelligence; mentoring of undergraduate students entering AI. [confirm: exact name of the organization]

TEACHING AND MENTORING

- Networking workshops taught at the Huawei training centre, Tehran, alongside the M.Sc. [confirm: scope and dates; this was a collaboration and teaching engagement, not employment]
- Instructor, Lab Instructor and Teaching Assistant, Queen’s University ECE, 2019–2023: MATLAB programming and mathematics (about 110 students per week); discrete-time signal processing in Python (about 90 students); laboratory sessions and student mentoring.
- Monthly seminar on continual learning, Ubiquitous Artificial Intelligence Laboratory (Prof. Il-Min Kim), Queen’s University ECE, 2019–2023: introduced and sustained the group’s engagement with the continual-learning literature.
- Networking instructor, Huawei training centre, Tehran, 2017–2019.
- Undergraduate mentoring, Ingenuity Lab, Queen’s University, 2019–2023.

PRESENTATIONS

- Autoencoder-based hybrid replay for class-incremental learning. ICML 2025, Vancouver, July 2025 (presenting author; supported by the TMU Faculty of Science travel award).
- Federated class-incremental learning: a hybrid approach using latent exemplars and data-free techniques. ICLR 2025, Singapore, April 2025 (presenting author).
- Task confusion and catastrophic forgetting in class-incremental learning. NeurIPS 2024, Vancouver, December 2024 (presenting author).
- A QGSA cluster head selection approach for hierarchical routing in EH-WSNs. ICSPIS 2018, Tehran (presenting author).
- Semi-annual research presentations to Huawei Technologies Canada, Ottawa, 2019–2023.
- Regular public talks on the impacts and opportunities of artificial intelligence, Ingenuity Lab, Queen’s University, 2019–2023.
- [fill: invited talks, seminars, guest lectures, if any]

TECHNICAL SKILLS

Machine learning. PyTorch, TensorFlow, NumPy, pandas, scikit-learn; timm and Hugging Face model zoos for pretrained backbones (ViT, DINOv2, CLIP, MAE, ResNet); experimental discipline built around pre-registration, seeded reproducibility and single-GPU budgets; Google Colab and workstation GPUs.

Theory. Learning theory for incremental and federated settings; concentration and margin bounds;

competitive analysis of online algorithms; convex and combinatorial optimization (integer programming).
Programming and data. Python, MATLAB, SQL, Linux and shell; LaTeX; data cleaning and exploratory-analysis pipelines for medical imaging and mass-spectrometry data.
Languages. Persian (native), English (fluent), Arabic (reading), French (elementary).

REFERENCES

Prof. Il-Min Kim

Professor and Head, Electrical and Computer Engineering
 Queen's University
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Prof. Sidney Givigi

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Prof. Guanghui (Richard) Wang

Professor, Department of Computer Science
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Prof. Saeed Sharifian

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